

Covariant Profile Geometry and the Central Complex Metric Channel

Flat ambient transport, induced Levi–Civita curvature, and the status of quaternionic components

Ing. David Jaroš

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Abstract

The free-profile GR candidate was initially written in a fixed profile frame with $E_\mu = \partial_\mu \Theta$. This note supplies the corresponding frame-covariant formulation without introducing a propagating connection. The profile carrier is treated as a flat complex Krein bundle with a flat, pairing-compatible ambient connection $\mathbb{D}$. Its tangential projection is the curved Levi–Civita connection of the induced real metric, while its normal projection is the second fundamental form. In a local parallel ambient frame $\mathbb{D} = \partial$, so the existing free-embedding and rank-ten proofs apply unchanged; their statements are frame invariant. The construction therefore does not reintroduce the single-section concurrent-vector obstruction.

We also prove an algebraic strengthening relevant to the full UBT field: for arbitrary biquaternions the $\sharp$ -symmetrised product is always central and complex-valued. Thus the natural symmetric extension is a central complex metric $\gamma_{\mu\nu} = g_{\mu\nu} + ih_{\mu\nu}$. A genuinely quaternion-valued component survives only in the antisymmetric bivector channel, not in the symmetric metric channel. Exact classical GR is the real Lorentzian branch $h_{\mu\nu} = 0$ (or a separately justified real readout).

Canonical status

This closes the *fixed ambient frame* objection inside the noncanonical profile architecture. It does not identify the flat ambient connection $\mathbb{D}$ with the currently canonical pointwise spin lift $D_\mu = \partial_\mu + \Omega_\mu(\cdot) + (\cdot)\Omega_\mu^\sharp$. Canonical promotion therefore requires an explicit three-way distinction:

$$\begin{aligned} \mathbb{D}_\mu & \text{ flat ambient/profile transport,} \\ \nabla_\mu & \text{ induced Levi–Civita transport,} \\ \Omega_\mu & \text{ spin lift of } \nabla_\mu. \end{aligned}$$

No new propagating field is required, because the ambient connection is flat and locally pure gauge.

1 The symmetric $\sharp$ channel is central on all of $\mathbb{B}$

Write arbitrary biquaternions as

$$q = z^0 \mathbf{1} + z^k \mathbf{e}_k, \quad r = w^0 \mathbf{1} + w^k \mathbf{e}_k, \quad z^a, w^a \in \mathbb{C},$$

with quaternion conjugation $q^\sharp = z^0 \mathbf{1} - z^k \mathbf{e}_k$.

Theorem 1.1 (Global centrality identity). *For every $q, r \in \mathbb{B} = \mathbb{C} \otimes \mathbb{H}$,*

$$\boxed{\frac{1}{2}(q^\sharp r + r^\sharp q) = \left(z^0 w^0 + \sum_{k=1}^3 z^k w^k \right) \mathbf{1} \in \mathbb{C} \mathbf{1}.} \quad (1)$$

The centrality does not require the real Lorentz slice. The Lorentz slice is needed to make the coefficient real with signature $(-, +, +, +)$.

Proof. Using $(a, \mathbf{u})(b, \mathbf{v}) = (ab - \mathbf{u} \cdot \mathbf{v}, a\mathbf{v} + b\mathbf{u} + \mathbf{u} \times \mathbf{v})$ and applying it to $q^\sharp = (z^0, -\mathbf{z})$, the vector parts of $q^\sharp r$ and $r^\sharp q$ cancel in the sum, while both scalar parts equal $z^0 w^0 + \mathbf{z} \cdot \mathbf{w}$. $\square$

Corollary 1.2 (Real Lorentz slice). *If $q = ix^0 \mathbf{1} + x^k \mathbf{e}_k$ and $r = iy^0 \mathbf{1} + y^k \mathbf{e}_k$ with real coefficients, then*

$$\frac{1}{2}(q^\sharp r + r^\sharp q) = (-x^0 y^0 + x^k y^k) \mathbf{1}.$$

Thus the active canonical metric is the real slice of a more general central complex bilinear.

Definition 1.3 (Central complex profile metric). *For arbitrary smooth $\mathbb{B}$ -valued profiles let*

$$\beta(U, V) \mathbf{1} := \frac{1}{L_\psi} \int_0^{L_\psi} \frac{1}{2} (U^\sharp V + V^\sharp U) d\psi, \quad (2)$$

$$\gamma_{\mu\nu}[\Theta] := \frac{1}{\mathcal{N}_0} \beta(E_\mu, E_\nu) = g_{\mu\nu} + i h_{\mu\nu}. \quad (3)$$

Here $g_{\mu\nu}$ and $h_{\mu\nu}$ are real symmetric tensors. The exact GR branch requires g nondegenerate Lorentzian and $h = 0$, unless a separate projection or dynamics for h is derived.

Proposition 1.4 (Where the quaternionic information goes). *The full averaged product decomposes uniquely as*

$$\frac{1}{\mathcal{N}_0 L_\psi} \int E_\mu^\sharp E_\nu d\psi = \gamma_{\mu\nu} \mathbf{1} + \Sigma_{\mu\nu}, \quad \Sigma_{\nu\mu} = -\Sigma_{\mu\nu}. \quad (4)$$

Therefore the $\sharp$ construction admits a central complex symmetric metric and a biquaternion-valued antisymmetric bivector, but no noncentral quaternion-valued symmetric metric.

Remark 1.5 (Hermitian alternatives). A symmetrised product based on the full Hermitian involution $\ddagger$ can be Hermitian and noncentral. Such an object may be useful as a generalized quadratic tensor, but it is not an ordinary spacetime metric: left/right inverses, determinant, index raising and Levi-Civita uniqueness become noncommutative questions. It must not be inserted into the GR closure under the name $g_{\mu\nu}$ without a separate geometric theory.

2 Flat ambient profile bundle

Let $\mathcal{V} \rightarrow M$ be the profile bundle with fiber $\mathcal{V}_\psi$ and complex bilinear form β . The relevant covariance is covariance under changes of the ambient profile frame.

Definition 2.1 (Ambient profile connection). *A connection $\mathbb{D}$ on $\mathcal{V}$ is admissible when*

$$\partial_\mu \beta(U, V) = \beta(\mathbb{D}_\mu U, V) + \beta(U, \mathbb{D}_\mu V), \quad (5)$$

$$[\mathbb{D}_\mu, \mathbb{D}_\nu] = 0. \quad (6)$$

It is therefore pairing-compatible and flat. Locally there is a parallel profile frame in which $\mathbb{D}_\mu = \partial_\mu$. In another frame it contains a pure-gauge connection one-form.

Under a local profile-frame isometry $U(x)$,

$$\Theta \mapsto U\Theta, \quad \mathbb{D}_\mu \mapsto U\mathbb{D}_\mu U^{-1}, \quad E_\mu := \mathbb{D}_\mu \Theta \mapsto U E_\mu.$$

The pointwise real Lorentz/Spin action is a subgroup of these pairing isometries; no preferred ambient basis is physical.

Proposition 2.2 (Gauge invariance of the complex and real metrics). *The tensors $\gamma_{\mu\nu}$, $g_{\mu\nu} = \text{Re } \gamma_{\mu\nu}$ and $h_{\mu\nu} = \text{Im } \gamma_{\mu\nu}$ are invariant under every admissible local ambient frame change.*

Proof. Pairing compatibility gives $\beta(U E_\mu, U E_\nu) = \beta(E_\mu, E_\nu)$. The spacetime indices are untouched, so the result follows componentwise. $\square$

3 The curved Levi–Civita connection is induced, not ambient

Restrict now to the real branch $h_{\mu\nu} = 0$ with nondegenerate Lorentzian g . Let

$$\mathcal{T}_x = \text{span}_{\mathbb{R}}\{E_0, E_1, E_2, E_3\} \subset \mathcal{V}_\psi$$

and let $P_\top, P_\perp$ be the orthogonal projections with respect to $H = \text{Re } \beta$.

Theorem 3.1 (Gauss decomposition and induced Levi–Civita connection). *There is a unique decomposition*

$$\boxed{\mathbb{D}_\mu E_\nu = \Gamma^\rho_{\mu\nu} E_\rho + B_{\mu\nu}, \quad B_{\mu\nu} \perp \mathcal{T}_x.} \quad (7)$$

For a flat ambient connection and $E_\mu = \mathbb{D}_\mu \Theta$:

1. $\Gamma^\rho_{\mu\nu} = \Gamma^\rho_{\nu\mu}$ and $B_{\mu\nu} = B_{\nu\mu}$;
2. Γ is metric compatible with g ;
3. hence Γ is exactly the Levi–Civita connection of g .

Proof. Nondegeneracy of the tangent Gram matrix gives the unique tangent/normal splitting. Flatness implies

$$\mathbb{D}_\mu E_\nu - \mathbb{D}_\nu E_\mu = [\mathbb{D}_\mu, \mathbb{D}_\nu] \Theta = 0,$$

so the tangent and normal parts are separately symmetric. Pairing compatibility and Eq. (7) give

$$\partial_\mu g_{\nu\lambda} = \Gamma^\rho_{\mu\nu} g_{\rho\lambda} + \Gamma^\rho_{\mu\lambda} g_{\nu\rho},$$

because the B terms are normal. Thus Γ is torsion free and metric compatible, which characterizes the Levi–Civita connection uniquely. $\square$

Corollary 3.2 (Gauss equation). *Flatness of $\mathbb{D}$ gives*

$$R_{\mu\nu\rho\sigma}[g] = \frac{1}{\mathcal{N}_0} (H(B_{\mu\rho}, B_{\nu\sigma}) - H(B_{\mu\sigma}, B_{\nu\rho})). \quad (8)$$

Thus a flat ambient/profile connection is fully compatible with a curved four-dimensional metric.

4 Covariant free rank and the concurrent-vector no-go

Theorem 4.1 (Covariant persistence of the free-profile completion). *In a local parallel ambient frame, $\mathbb{D} = \partial$ and the free-embedding lift of `gap_10r_free_fiber_embedding_completion.tex` applies verbatim. The induced metric, the ten normal vectors $B_{\mu\nu}$, their rank, and the Regge–Teitelboim stationarity equation are ambient-frame invariants. Therefore local representability and rank-ten vacuum closure persist in every admissible profile frame.*

Proof. Flatness supplies a local parallel frame. In that frame all formulas reduce to the already proved fixed-frame construction. An admissible frame change is an invertible isometry of the target fiber, so it preserves the tangent-normal split, linear independence of the ten $B_{\mu\nu}$, the induced metric and the Euler–Lagrange equation. Hence the result is frame independent. $\square$

Proposition 4.2 (Why the pointwise no-go does not return). *The concurrent-vector proof assumes that E_μ is a nondegenerate tetrad in a four-real-dimensional Lorentz carrier and that the same Levi–Civita spin connection acts on the field from which that tetrad is derived. In the profile architecture:*

1. E_μ lies in an ambient profile space containing at least a fourteen-dimensional free block;
2. $\mathbb{D}$ is the flat ambient connection;

3. the curved Levi–Civita connection is only the tangential projection of $\mathbb{D}$ along the embedded four-manifold.

There is therefore no profile-space inverse tetrad converting Θ into a single spacetime vector V with $\nabla_\mu V^\nu = \delta_\mu^\nu$.

5 Result and remaining obligations

Item	Status	Meaning
Centrality of symmetric $\sharp$ product	Closed	Holds on all $\mathbb{B}$; coefficients may be complex.
Classical real metric	Closed algebraically	Real Lorentz slice, or reality condition $h = 0$.
Quaternionic symmetric metric	Excluded in this channel	Noncentral information is the antisymmetric $\Sigma_{\mu\nu}$.
Profile-frame covariance	Closed conditionally	Use a flat β -compatible ambient connection $\mathbb{D}$.
Induced Levi–Civita geometry	Closed	Tangential projection of $\mathbb{D}$; Gauss equation follows from ambient flatness.
Free rank under covariantization	Closed conditionally	Equivalent to the fixed-frame proof in a local parallel gauge.
Identification with old canonical D_μ	Open architectural revision	The old pointwise spin lift must not be conflated with $\mathbb{D}$.
Action origin and mode selection	Open	Unchanged by this note.

The correct candidate architecture is therefore

$$\boxed{\Theta \xrightarrow{\mathbb{D}} E_\mu \xrightarrow{\beta} \gamma_{\mu\nu} = g_{\mu\nu} + ih_{\mu\nu}, \quad \mathbb{D}_\mu E_\nu = \Gamma^\rho_{\mu\nu} E_\rho + B_{\mu\nu}.} \quad (9)$$

Exact GR uses the real branch and the induced Levi–Civita connection. The complex symmetric tensor $h_{\mu\nu}$ and the biquaternionic antisymmetric sector $\Sigma_{\mu\nu}$ may be retained as candidate non-GR sectors, but neither has a physical interpretation until its action, constraints and couplings are derived.